

Mathematics for Data Science – True/False Review

For each statement, explain whether it is **true** or **false**.

Lecture 1: Vector spaces and linear maps

1. A subset S of a vector space V is a subspace if and only if it contains the zero vector and is closed under addition and scalar multiplication.
2. Every affine line in $\mathbb{R}^n$ is a vector subspace of $\mathbb{R}^n$.
3. If a list of vectors spans a vector space, then the list is necessarily linearly independent.
4. A linear map $L : U \rightarrow V$ is injective if and only if $\ker(L) = \{0_U\}$.

Lecture 2: Matrices and linear systems

1. If $A \in \mathbb{R}^{m \times n}$ and $x \in \mathbb{R}^n$, then Ax is defined and belongs to $\mathbb{R}^m$.
2. The sum $A + B$ is defined for any two real matrices, even when their dimensions are different.
3. For $A \in \mathbb{R}^{m \times n}$, the rank–nullity theorem is

$$n = \text{rank}(A) + \dim(\ker(A)).$$

4. The map $x \mapsto Wx + b$ is linear for every matrix W and every vector b .

Lecture 3: Determinants

1. The determinant is defined for every rectangular matrix.
2. For square matrices A and B of the same size, $\det(AB) = \det(A)\det(B)$.
3. Exchanging two rows of a matrix leaves its determinant unchanged.
4. A square matrix A is invertible if and only if $\det(A) \neq 0$.

Lecture 4: Eigenvalues and diagonalization

1. An eigenvector must be nonzero and satisfy $Av = \lambda v$ for some scalar λ .
2. The zero vector is an eigenvector for every eigenvalue of a linear map.
3. If an endomorphism has n pairwise-distinct eigenvalues on an n -dimensional space, then it is diagonalizable.
4. For a finite-dimensional endomorphism $f : E \mapsto E$, the characteristic polynomial has degree $\dim(E) - 1$.

Lecture 5: Normed and inner product spaces

1. An inner product on a real vector space is positive-definite: $\langle x, x \rangle \geq 0$, with equality if and only if $x = 0$.
2. A metric space must also be a vector space.
3. A norm satisfies $\|\alpha x\| = |\alpha| \|x\|$ for every real scalar α .
4. In a finite-dimensional vector space, any two norms are equivalent: there are constants $c, C > 0$ such that

$$c\|x\|_A \leq \|x\|_B \leq C\|x\|_A$$

for every vector x .

Answer key

Lecture	1	2	3	4
1	True	False	False	True
2	True	False	True	False
3	False	True	False	True
4	True	False	True	False
5	True	False	False	True

NB: **Totals:** 10 True and 10 False.

Justifications

1. **True.** These are precisely the three subspace conditions: $0 \in S$, closure under addition, and closure under scalar multiplication.
2. **False.** A vector subspace must contain the origin. The correct statement is that an affine line is a vector subspace only when it passes through the origin. For example, $\{(1, t) : t \in \mathbb{R}\} \subset \mathbb{R}^2$ does not contain $(0, 0)$.
3. **False.** Spanning means that every vector can be expressed as a linear combination; linear independence additionally requires that this representation be unique. For example, $(1, 0), (0, 1), (1, 1)$ spans $\mathbb{R}^2$ but is linearly dependent.
4. **True.** If $L(u) = L(v)$, then $L(u - v) = 0$, so injectivity is equivalent to the kernel containing only the zero vector.
5. **True.** Matrix multiplication has compatible dimensions: an $m \times n$ matrix times an n -vector produces an m -vector.
6. **False.** Matrix addition requires equal dimensions. For example, a 2×2 matrix cannot be added to a 2×3 matrix.
7. **True.** The rank-nullity theorem says that the dimension of the domain, here n , equals rank plus nullity.
8. **False.** The correct statement is that $x \mapsto Wx + b$ is affine; it is linear only when $b = 0$ (or in the degenerate case where the vector space contains only zero).
9. **False.** Determinants are defined only for square matrices, not rectangular matrices.
10. **True.** The determinant is multiplicative for square matrices of compatible, equal size.
11. **False.** Exchanging two rows multiplies the determinant by -1 .
12. **True.** A square matrix is invertible exactly when its determinant is nonzero.
13. **True.** By definition, an eigenvector is nonzero and satisfies $Av = \lambda v$.
14. **False.** The zero vector is excluded from the definition because it satisfies $A0 = \lambda 0$ for every scalar λ , so it would not identify a particular eigenvalue.
15. **True.** Eigenvectors associated with pairwise-distinct eigenvalues are linearly independent. n such vectors therefore form a basis of an n -dimensional space, which diagonalizes the endomorphism.

- 16. False.** The characteristic polynomial of an endomorphism on an n -dimensional space has degree $n = \dim(E)$.
- 17. True.** Positive-definiteness is one of the defining axioms of a real inner product.
- 18. False.** A metric space is just a set equipped with a distance satisfying positivity, symmetry, and the triangle inequality. For example, $([0, 1], d)$ with $d(x, y) = |x - y|$ is a metric space but not a vector space over $\mathbb{R}$.
- 19. False.** Absolute homogeneity is $\|\alpha x\| = |\alpha|\|x\|$. The stated formula fails for negative α .
- 20. True.** This is the equivalence-of-norms result for finite-dimensional vector spaces; the constants depend on the two norms, not on x .